# PALEOMAGNETISM OF THE CAMBRIAN ROCKS OF THE GREAT VALLEY OF EAST CENTRAL PENNSYLVANIA: FOLD TEST CONSTRAINTS ON THE AGE OF MAGNETIZATION

Richard J. Stead* and Kenneth P. Kodama

Department of Geological Sciences, Lehigh University,
Bethlehem, PA 18015

Abstract. A paleomagnetic study of the upper Cambrian Allentown Dolomite and the Middle Cambrian Leithsville Formation from east central Pennsylvania shows that these carbonate rocks carry a secondary magnetization with south declinations and shallow inclinations. These directions are isolated by Zijderveld [1967] analysis of thermal and alternating field demagnetization data. The Allentown's magnetization fails both an Alleghenian age fold test and a Taconic age fold test. The Leithsville data fail an Alleghenian age fold test. Comparison of the in situ magnetization directions with the North America APW path suggest that these formations were remagnetized in the Permian. Isothermal remanent magnetization (IRM) acquisition data, Lowrie-Fuller tests and thermal demagnetization of a saturation IRM suggest that the magnetic carrier is multidomain magnetite. The magnetite may be (1) authigenic magnetite or (2) detrital magnetite which had its magnetization reset thermally as suggested by the conodont alteration indices of Ordovician rocks in the region. Preliminary work on the Hardyston Quartzite suggests that it may carry a primary Cambrian magnetization.

## Introduction

One of the basic problems confronting paleomagnetists in North America is the interpretation of the large swath of Cambrian paleomagnetic poles for the craton from approximately 0°N lat., 160°E long. to 50°N lat., 100°E long. [Irving, 1979; Watts et al., 1980; Seguin et al., 1981; Dankers and LaPointe, 1981]. Watts et al. [1980] have suggested that the large arc of paleomagnetic poles is due to rapid apparent polar wander (apw) for North America during the Cambrian and that the data should be interpreted as a loop or hairpin in the apw path. Watts et al., [1980] point out that there is a general progression in age from Late Precambrian and Early Cambrian poles at 0°N., 160°E. to Middle Cambrian poles at approximately 50°N., 100°E. and then back to 0°N. 160°E. for upper Cambrian data. They note, however, that the data distribution could also be caused by partial to complete remagnetization of Cambrian strata during the Late Paleozoic [Creer, 1968]. The lack of adequate fold tests to constrain the age of magnetization of North American Cambrian rocks has confounded workers trying to determine the reality of the loop.

The thick ( ∿850 m. [Drake, 1969]) Cambrian section in the Lehigh Valley of east central Pennsylvania affords an opportunity to test the Watts et al., [1980] Cambrian apw loop. The rocks have been extensively folded during the Taconic and Alleghenian orogenies allowing the age of magnetization to be constrained by Graham's [1949] fold test.

## Geology

The Cambrian rocks of the Lehigh Valley consist of three formations. In stratigraphic sequence from bottom to top these are the Hardyston Quartzite, the Leithsville Formation and the Allentown Dolomite (Figure 1). Sparse fossil occurrence in these rocks limits the accuracy with which they are dated.

The Hardyston Quartzite of Early Cambrian age is a thin ( ∿30 m.), poorly exposed unit of arkose, feldspathic sandstone, orthoquartzite, quartz pebble conglomerate, silty shale and jasper [Drake, 1969; Aaron, 1969]. It is overlain by the Leithsville Formation, a shallow platform carbonate approximately 300 m. thick. The Leithsville is composed of nearly pure dolomite with minor phyllites and dolomite-cemented sands. The only known fossils in the Leithsville are possible Hyolithellus micans (Early Cambrian?) from a quarry near Califon, New Jersey [Drake, 1969]. However, by stratigraphic arguments the Leithsville is considered to be Middle Cambrian in age [Willard, 1961; Drake,

*Now at Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, CA 91125.



Copyright American Geophysical Union

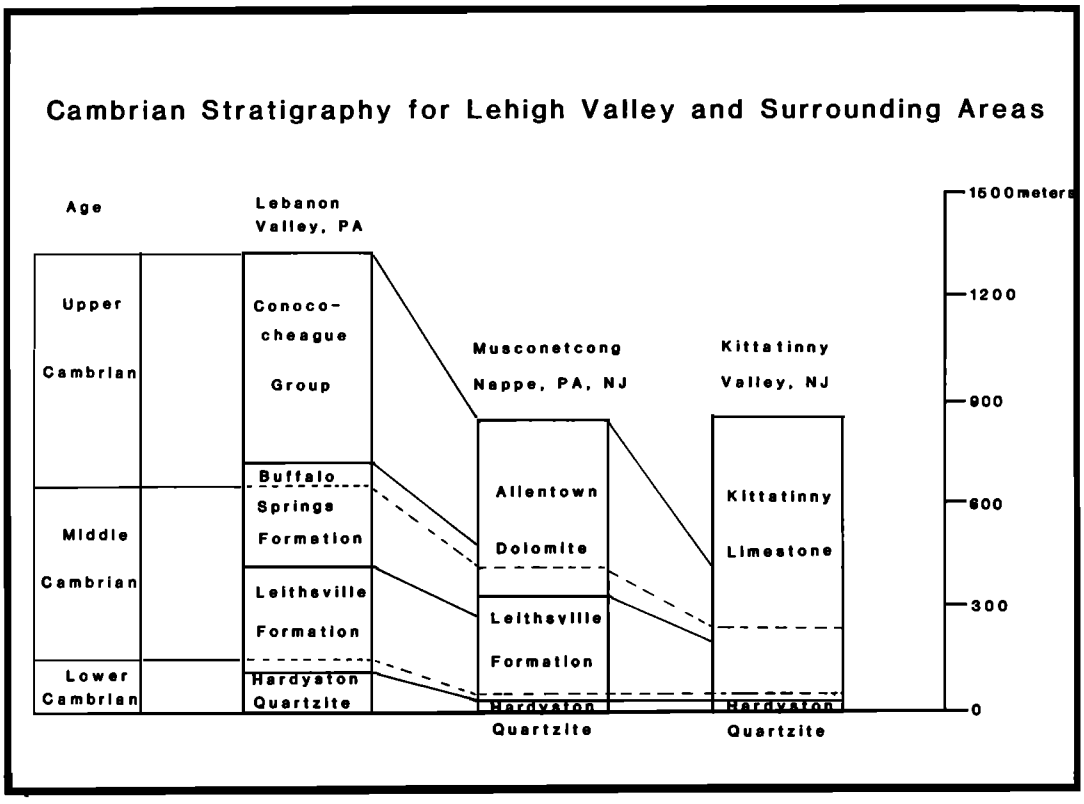



Fig. 1. Cambrian Stratigraphy of the Lehigh Valley and neighboring regions to the east and west illustrating thickness, probable ages and correlations. After Drake (1969).

1969]. The contacts of the Leithsville with both the overlying Allentown and the underlying Hardyston are gradational.

The Allentown Dolomite is a shallow subtidal to supratidal carbonate which correlates with the Conococheague group to the west and part of the lower Kittatinny limestone to the east [Drake, 1969] (Figure 1). The Allentown is nearly pure dolomite, completely recrystallized, and has abundant oolites and stromatolites. The unit is 520 m. thick and is considered to be Late Cambrian in age [Drake, 1969].

Since their deposition, these rocks have been subjected to a complex tectonic history. There are two major orogenies recognized in the region; the Taconic and the Alleghenian. The Taconic orogeny occurred in the Late Ordovician, producing large nappes and thrusting the rocks toward the northwest [Drake, 1969 and 1978]. For this reason the Cambrian units of the Lehigh Valley are allochthonous, however, A. A. Drake, P. B. Myers, and G. G. Lash (personal communications) all estimate that the amount of displacement is on the order of 100 km, which would displace the paleopole by less than 1°, assuming that there have been no local rotations. The Permian Alleghenian orogeny produced an extensive series of relatively open, symmetric anticlines and synclines. These symmetric folds refold the limbs of the Taconic nappes and can be related to cleavage in the Ordovician Martinsburg Formation to the north [Drake, 1969]. Epstein and Epstein [1967, 1969] have shown that the Martinsburg cleavage is Late Paleozoic in age and was caused by the Appalachian (Alleghenian) orogeny.

Most of the sampling sites for this study are within the Musconetcong Nappe [Drake, 1969; 1978]. The axis of this nappe is poorly determined but it trends roughly northeast-southwest due to its formation by northwestward thrusting in the Taconic. Sites A and L are located in the upper, right-side-up limb of the Taconic Musconetcong Nappe (Figure 2). Sites Z, DA, DL, S, and C are in the lower, inverted limb of the nappe. However, an Alleghenian age synform and antiform refold the lower limb of the Musconetcong Nappe further complicating the picture. Site DA, on the southern limb of the antiform and site Z on the northern limb of the synform (Figure 2) are inverted since they are also on the lower limb of the Musconetcong Nappe (Figure 2). However, sites S, C, and DL which are further to the west, have been re-inverted by overturning of the limbs of the antiform and the synform. These sites are now right-side-up due to Alleghenian age folding (Figure 2). These structural relationships will



Copyright American Geophysical Union

10.1029/GD012p0120. Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

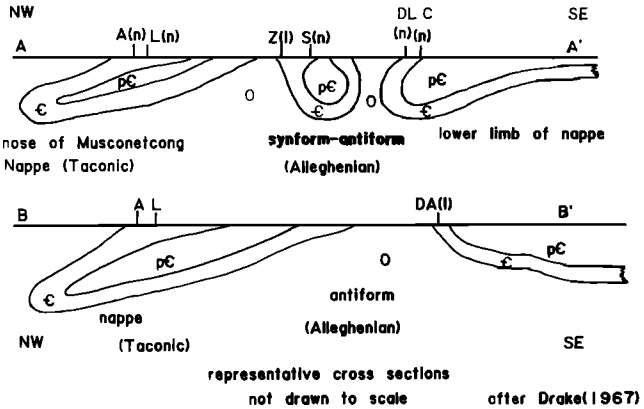



Fig. 2. Representative cross sections NE-SW across the structural grain of the Lehigh Valley. Sites A and L, which are much further to the SW, are projected along strike on to the profiles (see Fig. 3). The two sections show the comlicated structure of the Alleghenian antiform and synform along strike. n = right side up bedding; i = inverted bedding.

help to constrain the age of magnetization in the Allentown and Leithsville Formations.

## Methods

All samples from the Leithsville and Allentown Formations were collected with a portable, gasoline powered, diamond coring drill and were oriented in the field with a Brunton compass. Nineteen cores were drilled at three sites in the Leithsville Formation and 45 cores were sampled at 5 sites in the Allentown Dolomite. These sites were selected over an area of about 100 $km^2$ to minimize local effects (Figure 3). Fifteen specimens were drilled from five hand samples collected at two sites in the Hardyston Quartzite.

All samples were measured on ScT cryogenic magnetometers (3 axis-University of Pittsburgh; 2 axis-Lamont-Doherty Geological Observatory). Most cores were cut into two specimens, one being subjected to progressive thermal demagnetization using a Schonstedt TSD-1 and the other subjected to progressive alternating field demagnetization with either a 2-axis tumbling, Inductrol-controlled apparatus or a Schonstedt GSD-1.

Rock magnetic studies were conducted with a digitized, fluxgate slow spinner magnetometer, a 2-axis tumbling, Inductrol-controlled, 60 cps alternating field demagnetizer, a thermal demagnetizer, and an electromagnet capable of 1.3 T. DC fields. Five representative dolomite samples from the Leithsville and Allentown Formations were progressively saturated in peak DC magnetic fields up to 0.4 T. Thermal demagnetization of a saturation isothermal remanent magnetization (SIRM) was conducted as well as alternating field demagnetization of an SIRM and an anhysteretic remanent magnetization (ARM) (DC field = 0.1 mT; peak AC field = 150 mT).

## Results

### Leithsville Formation

Zijderveld [1967] analysis of demagnetization data from samples of the Leithsville Formation suggests that a characteristic magnetization has been isolated (Figure 4). Both thermal and alternating field demagnetization yield a south and shallow characteristic magnetization for most of the samples before the data are corrected for bedding tilt. Approximately 20% of the Leithsville samples have north and shallow magnetizations; however these directions have been inverted through the origin to calculate site means which have south declinations. Both negative and positive inclinations are isolated for south and north declinations (Table 1). The data show a relatively univectorial decay toward the origin after removal of steep, positive inclination component at alternating fields of 10 mT - 20 mT and temperatures between 150°C-200°C (Figure 4).

As noted before the structural corrections for the bedding tilt of sites DL, S & C (Figure 2) are not straightforward. All three sites lie on the inverted limb of the Taconic Musconetcong Nappe; however, the sites are right-side-up due to reinversion on the limbs of an Alleghenian synform and antiform [Drake, 1967] (Figure 2). A two stage bedding correction seems reasonable. First the magnetic directions will be unfolded around the Alleghenian antiform and synform. This will give bedding which is inverted by the Taconic Musconetcong Nappe. Since the bedding strike of sites C, DL, and S is nearly parallel to the antiform and synform axes and the fold plunge is very slight ( $<10°$ P. B. Myers, personal communication) simple rotations ( $<90°$) around bedding strike were made to bring the beds into their post-Taconic pre-Allehenian orientations (Figure 5). Clearly the data suggest that the Leithsville magnetization fails an Alleghenian fold test and is post-Alleghenian in age. A subsequent inversion of the beds to unfold the Taconic Musconetcong Nappe is, therefore, unnecessary. Even if these sites are not rotated to their post-Taconic/pre-Alleghenian inverted orientation but are simply rotated around bedding strike by the amount of bedding dip ( $<90°$) to a horizontal right-side-up position the precision parameter, K, decreases from 94.6 to 63.6 (Figure 5) suggesting, again, failure of a regional tilt test.

### Allentown Dolomite

Zijderveld [1967] diagrams for the Allentown Formation (Figure 6) show demagnetization behavior similar to that observed for the



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

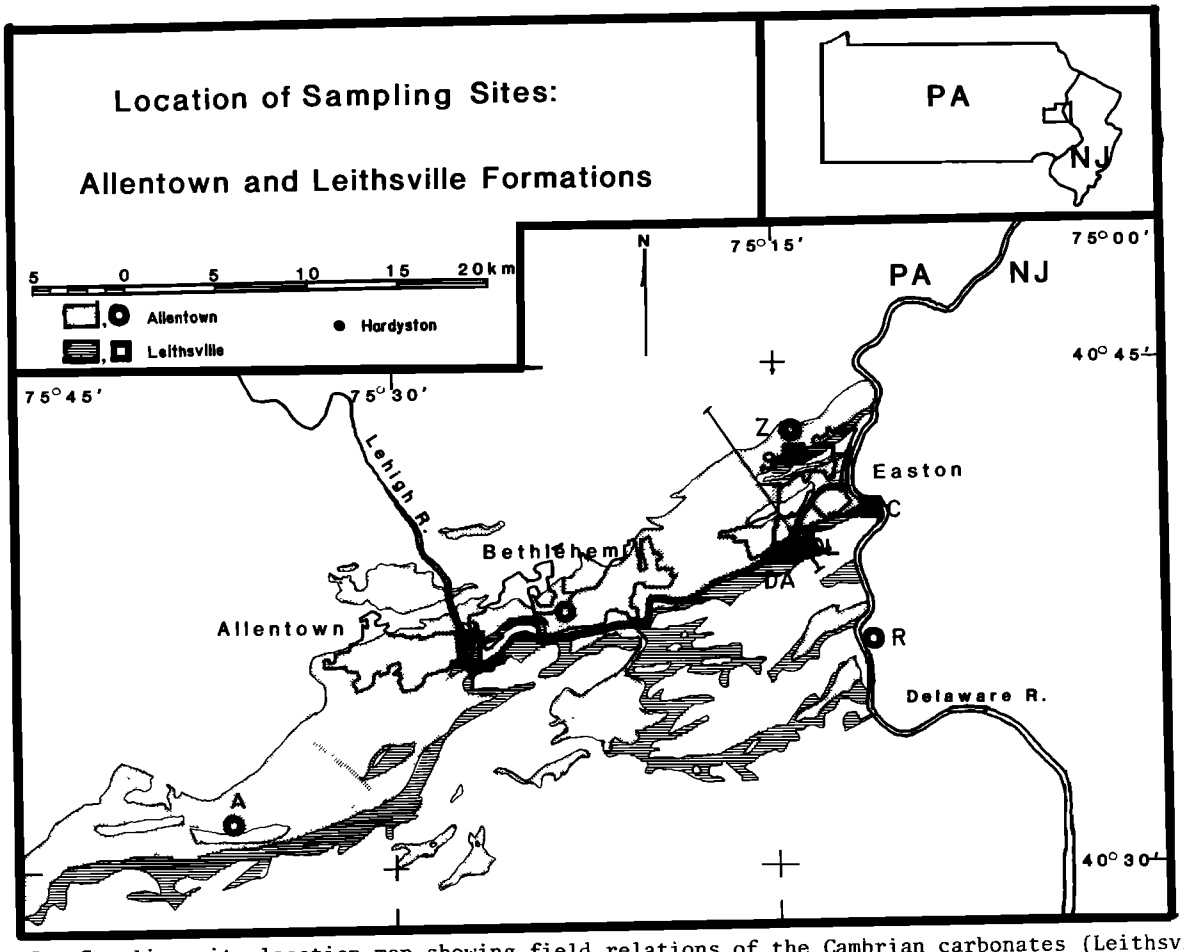



Fig. 3. Sampling site location map showing field relations of the Cambrian carbonates (Leithsville Formation and Allentown Dolomite) and the Hardyston Quartzite. Based on the Geological Map of Pennsylvania, 1980. NW-SE line shows approximate location of the sections in Fig. 2.

Leithsville Formation. The characteristic directions isolated by both thermal and alternating field demagnetization are in most cases south directed with either shallow positive or negative inclinations (Table 1). Only 4% of the samples had north-directed declinations. As in the Leithsville, the Allentown magnetizations decayed univectorially toward the origin after a steep secondary component was removed at low temperatures ( 200°C) and alternating fields ( <20mT) (Figure 6).

Two of the five sites in the Allentown Dolomite will be discussed in detail in the next section since at these sites Taconic (Site R) and Alleghenian (site L) fold tests were conducted. Of the remaining three sites, two (sites DA and Z) are located on the overturned limb of the Musconetcong Nappe (see Figure 2). The evidence for these sites' inversions come from stromatolite orientation and the structural interpretation of Drake [1967]. Site A is located on the upper, right-side-up limb of the Musconetcong Nappe. When these site directions are rotated about bedding strike for this slightly plunging fold the directions diverge suggesting a failure of a Taconic age fold test by the Allentown magnetization (Figure 7). The implications of this fold test will be discussed in more detail in the following section.

## Hardyston Quartzite

Only two sites (15 specimens/5 hand samples) were collected from the Early Cambrian Hardyston Quartzite. Although these specimens were pink in color most could only be thermally demagnetized to 580°C since they became too weak to measure on the cryogenic magnetometer (∿.06mA/m) or their magnetizations were very unstable during measurement after demagnetization at steps 580°C. Thermal demagnetization failed to isolate a characteristic magnetization for 3 specimens since the orthogonal projection end points did not trend toward the origin with increasing temperature. The characteristic magnetizations isolated for the remaining twelve specimens are



Copyright American Geophysical Union

10.1029/GD012p0120. Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

quite scattered before correction for bedding tilt (Figure 8).

A bedding tilt correction for the Hardyston is not a trivial matter. Bedding can only be recognized by looking for small conglomerate layers in the hand samples from which the specimens had been drilled. These features were identified on two intersecting faces of a hand sample and the true orientation of the bedding was calculated from the apparent bedding dips on the hand sample faces. The validity of this method is very dependent on our ability to correctly identify the conglomerate layers; not a straightforward task. When bedding corrections are made using these bedding orientations there is an apparent clustering of the data (Figure 8, Table 1). The seven specimen magnetizations which point approximately up and toward the N-NE represent 2 hand samples from one site and one hand sample from the other site (about 1 km away). The mean direction calculated from these specimens, using a hierarchical averaging scheme, is I= -52.0°, D= 25.7°, K= 9.2, $\alpha_{95}$= 26.7°. This corresponds to a paleolatitude of -32.6° which, when the large errors are considered, is consistent with this site's location in the Cambrian-Ordovician (500 M.Y. B.P.) [Irving, 1979; Watts, et al., 1980] or the Early Cambrian [Watts, et al., 1980]. However, its declination is approximately 90° clockwise from the expected Cambrian declination [Irving, 1979; Watts et al., 1980]. The Hardyston magnetization could also be a Silurian overprint since North America moved to higher latitudes during this time [Van der Voo et al., 1983]. The Hardyston declination would still be rotated clockwise with respect to a Silurian direction, but by a smaller amount ($\sim$45°).

## Fold Tests

In order to better constrain the age of the Allentown magnetization fold tests [Graham, 1949] of Taconic and Alleghenian age were conducted. An Alleghenian age anticline (A. A. Drake, personal communication) at Lerch's Quarry, Bethlehem, PA. is located in the upper right-side-up limb of the Musconetcong Nappe (Site L, Figure 2). Fifteen cores were drilled on both limbs of the symmetric fold. These samples were cut into 24 specimens which were thermal and alternating field demagnetized. Four specimens suffered from a steep, downward present field overprint which could not be removed by demagnetization. Twenty specimens representing 13 samples from both limbs of the fold give a mean direction of I=2.9°, D=169.5°, K=13.7, $\alpha_{95}$=10.5° before unfolding. This magnetization fails the Alleghenian fold test at the 99% level ($K2/K_1$=0.3205, N=13 [McElhinny, 1964] since the directions disperse to a K=4.4 after unfolding (Figure 9).

Thirteen cores were also drilled in the Allentown Dolomite at an overturned antiform at

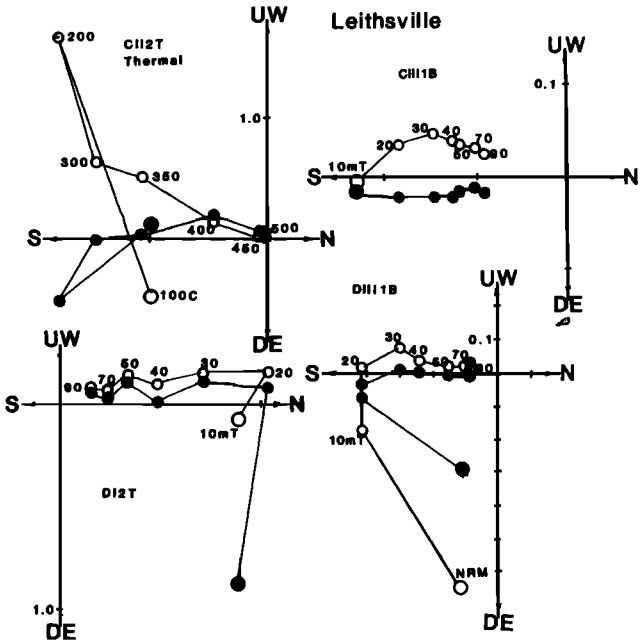



Fig. 4. Examples of Zijderveld (1967) diagrams for selected samples of the Leithsville Formation before tilt correction. Sample CII2T was treated thermally while the others received alternating field demagnetization. Sample DI2T shows a normal polarity sample. Solid circles indicate the horizontal component, and open circles indicate the vertical component. Units of the coordinate axes are mA/m.

Riegelsville, New Jersey which is Taconic in age (A. A. Drake, personal communication).

In light of the Allentown magnetization's failure of an Alleghenian fold test, it was not expected that it would pass a Taconic fold test. In addition to this expectation only seven of the thirteen samples had orthogonal projection end points which trended toward the origin with demagnetization. The magnetization which was isolated was plagued by a normal polarity present day overprint (Figure 9). As would be expected the overprinted magnetization failed the Taconic fold test.

In summary, it is apparent that the shallow south magnetizations observed in the Leithsville and the Allentown formations are not primary and represent a secondary magnetization which is younger than the Late Paleozoic Alleghenian orogeny.

## Rock Magnetic Results

The IRM acquisition data show that the dolomite samples become saturated in direct magnetic fields of 0.3 T and less (Figure 10). Plots of the decrease in intensity of NRM, ARM, and SIRM with alternating field demagnetization are normalized to the 10 mT demagnetization step to



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Site Summaries

| Site | Inverted | N†† | Before Tilt Correction Decl. | Incl. | K | α95 | After Tilt Correction Decl. | Incl. | K | 95 | Bedding Strike | Dip |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Allentown Dolomite | | | | | | |
| A | No | 4/4 | 179.6° | 3.7° | 35.8 | 8.8° | 179.3° | 1.0° | 35.8 | 8.8° | N28°E | 8.5°SE |
| DA | Yes | 4/7 | 172.5° | 15.8° | 217.2 | 3.9° | 11.5° | 16.0° | 157.1 | 4.6°+ | N89°W | 33.5°S |
| L* | No | 13/15 | 169.5° | -2.9° | 13.7 | 10.5° | 176.0° | -0.8° | 4.4 | 18.5° | * | |
| R** | Yes | 7/13 | 329.1° | 80.5° | 39.8 | 8.4° | | | | | | |
| Z | Yes | 4/6 | 171.9° | 2.6° | 60.7 | 6.8° | 268.1° | 26.5° | 60.3 | 6.8°+ | N43°E | 28.5°SE |
| | | | | | | Leithsville Formation | | | | | | |
| C | No | 4/7 | 173.7° | -13.1° | 101.5 | 5.7° | 253.8° | -7.0° | 104.5 | 5.6°+ | N33°E | 33°NW |
| DL | No | 5/6 | 174.9° | 2.2° | 47.8 | 7.7° | 11.5° | -29.0° | 53.5 | 7.2°+ | N87°W | 23.5°N |
| S | No | 6/6 | 176.2° | -10.8° | 87.1 | 6.1° | 311.5° | -37.0° | 70.9 | 6.8°+ | S66°W | 48.5 NW |
| Site | Inverted | N††† | Decl. | Incl. | K | α95 | Decl. | Incl. | K | 95 | Strike | Dip |
| | | | | | | Hardyston Quartzite | | | | | | |
| 1 & 2 | No | 3/3 | 3.8° | -74.3° | 3.1 | 30.0° | 25.7° | -52.0° | 9.2 | 26.7° | | |

| Formation | Sites | Decl. | Incl. | K | 95 | P lat | P long | M | P |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Formation Paleomagnetic Poles | | | | | |
| Allentown BTC | L,A,Z,DA | 173.4° | 4.8° | 81.3 | 7.8° | 46.6° | 114.4° | 7.8° | 3.9° |
| Leithsville | C,DL,S | 174.9° | -7.2° | 94.6 | 8.3° | 52.60° | 113.2° | 8.3° | 4.9° |

[1]Note: Reversed polarity directions given with equivalent North Paleomagnetic Pole
[2]* Alleghenian Fold Test at this site (north limb S78°E 25°N; south limb N65°E 41°S)
[3]** Taconic Fold Test at this site (Note: present day overprint BTC)
[4]† ATC is site corrected to pre-Alleghenian (post-Taconic orientation (see text))
[5]†† Number of samples accepted to the total number taken on the basis of Zijderveld (1967) analysis
[6]††† 7 specimens from 3 hard samples from 2 sites



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

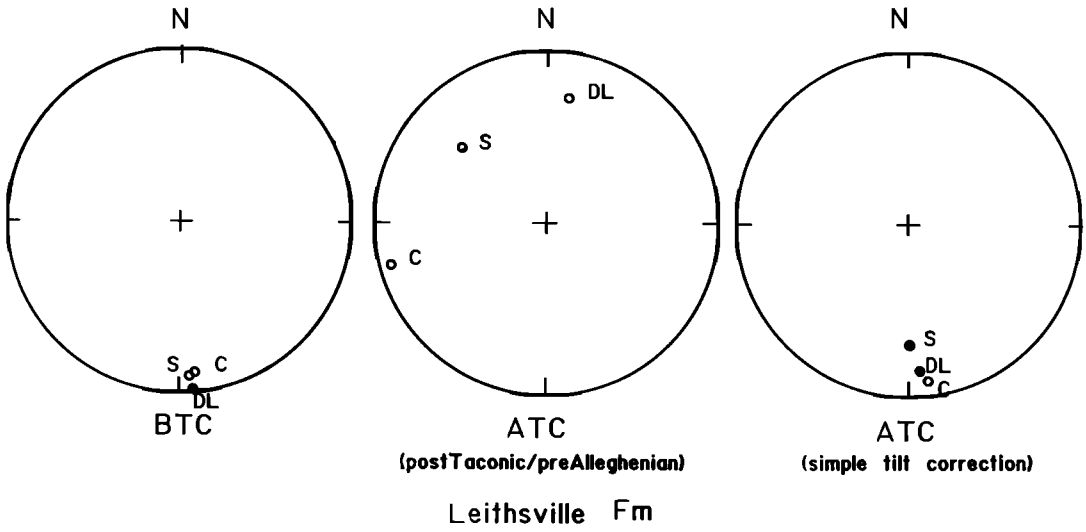



Fig. 5. Equal area projections showing the before tilt correction (BTC)site mean directions for the Leithsville Formation. The middle stereonet shows the post-Taconic/pre-Alleghenian tilt correction for these sites. The right stereonet shows a more conservative bedding tilt correction. In both cases the Leithsville fails an Alleghenian fold test. open circle = upper hemisphere; solid circle = lower hemisphere.

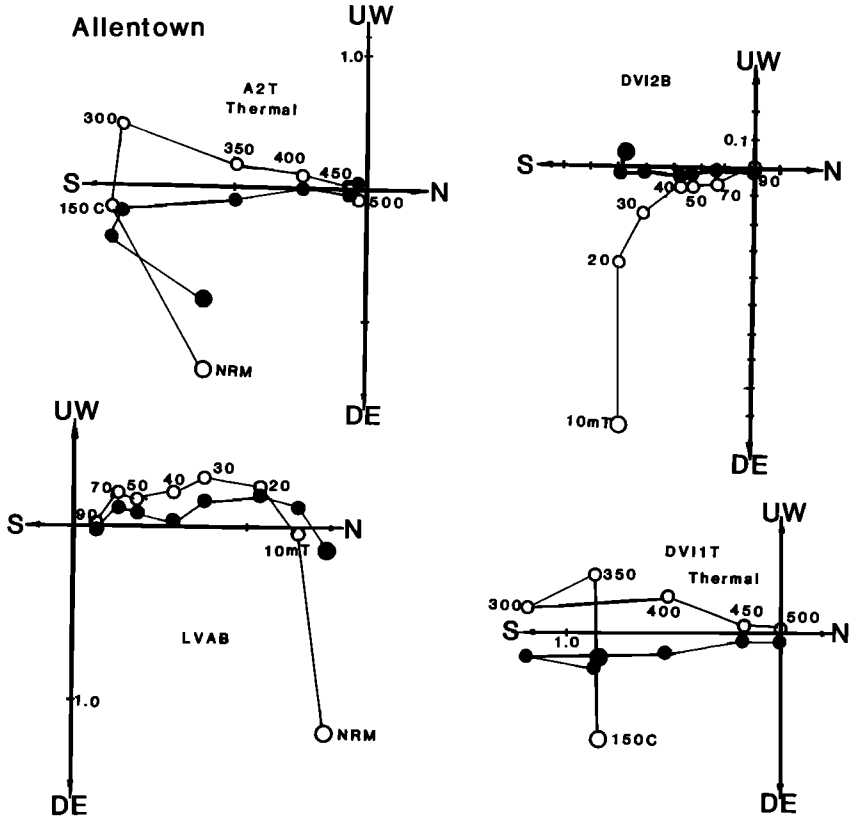



Fig. 6. Examples of Zijderveld (1967) diagrams for selected samples of the Allentown Dolomite. Samples A2T and DVI1T were treated thermally, while DVI2B and LVAB received alternating field demagnetization. Sample LVAB represents a normal polarity. Solid circles are horizontal components, open circles are vertical. Units are mA/m.



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

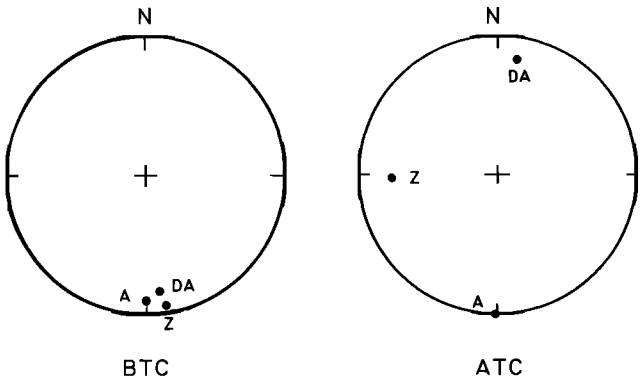



Fig. 7. Before (BTC) and after (ATC) tilt correction for the Allentown Dolomite sites. Site DA and Z have been inverted by Paleozoic folding. When these sites (Z and DA) are rotated around bedding strike to right-side-up positions the formation scatter increases markedly. Symbols the same as Fig. 5.

minimize the effects of the viscous magnetization acquired by the NRM [Lowrie and Fuller, 1971] (Figure 10). These data show that SIRM is more resistive to af demagnetization than NRM and ARM and that the NRM has approximately the same coercivity characteristics as the ARM and SIRM. Thermal demagnetization of an SIRM indicates a maximum unblocking temperature of about 500°C to 550°C, similar to the maximum unblocking temperatures of the NRM (Figure 10).

The rock magnetic experiments all indicate that magnetite is the predominate magnetic carrier in these carbonates. The IRM acquisition data show coercivities in the range expected for magnetite and much less than those expected for goethite or hematite [Lowrie and Heller, 1982]. When the af demagnetization behavior of NRM, ARM, and SIRM were compared, the alternating field demagnetization data were normalized to 10 mT because of the viscous component seen to be removed from the NRM in the Zijderveld [1967] diagrams. These data suggest that the NRM is similar in demagnetization character to ARM and SIRM and that the remanence is carried by multidomain grains [Lowrie and Fuller, 1971; Johnson et al., 1975]. Although it may seem surprising that a carbonate

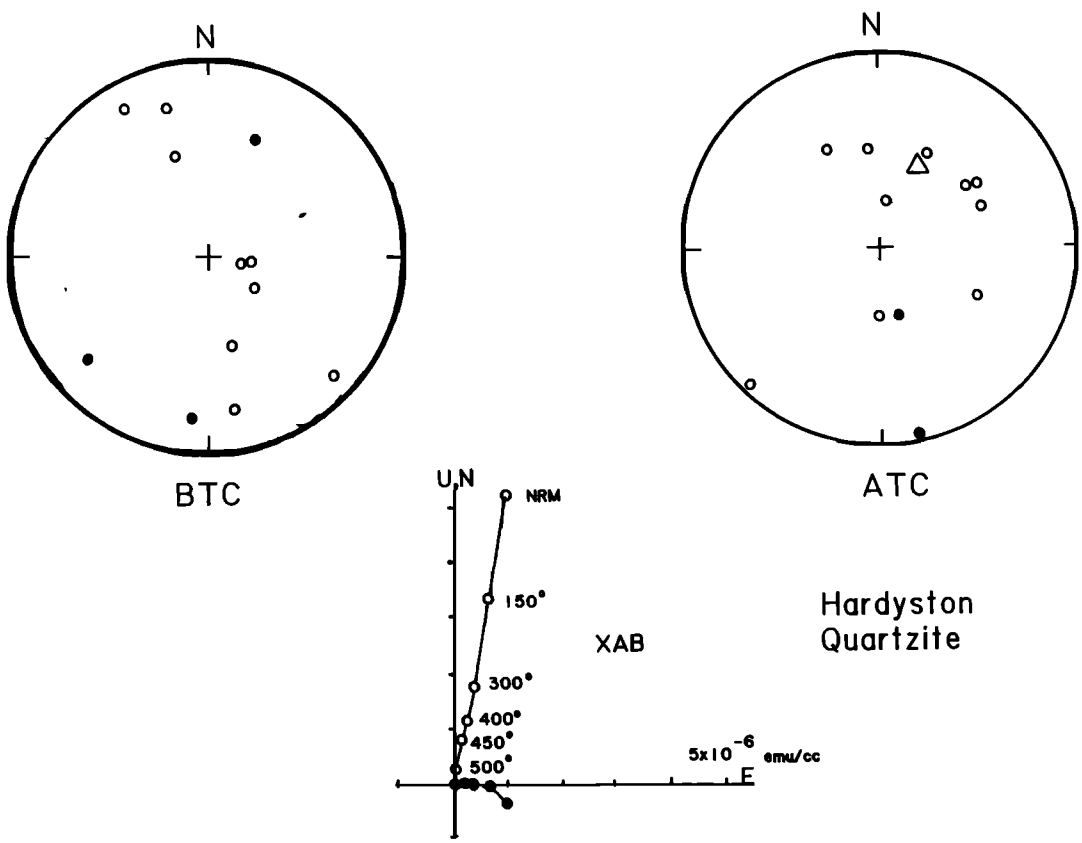



Fig. 8. Hardyston Quartzite individual specimen directions for 5 hand samples from 2 sites before tilt correction (BTC) and after tilt correction (ATC). Triangle shows mean direction for the seven specimens from 3 hand samples (from 2 sites) which print up and roughtly NNE.



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

rock contains multidomain magnetite, Lowrie and Heller [1982] point out that the Ordovician Trenton limestone [McElhinny and Opdyke, 1973], the Devonian Onondaga limestone [Kent, 1979], and the Carboniferous Pendleside limestone [Turner, 1975] all contain multidomain magnetic grains. The thermal demagnetization data rule out hematite as an important magnetic carrier in these rocks. Both the NRM and SIRM thermal demagnetization curves show maximum unblocking temperatures well below those commonly associated with hematite. The increase in intensity with thermal demagnetization of NRM at temperatures up to 300°C may be due to removal of a viscous secondary magnetization which is nearly antiparallel to the characteristic magnetization.

## Discussion

A reconnaissance study of the Hardyston Quartzite isolated a characteristic magnetization

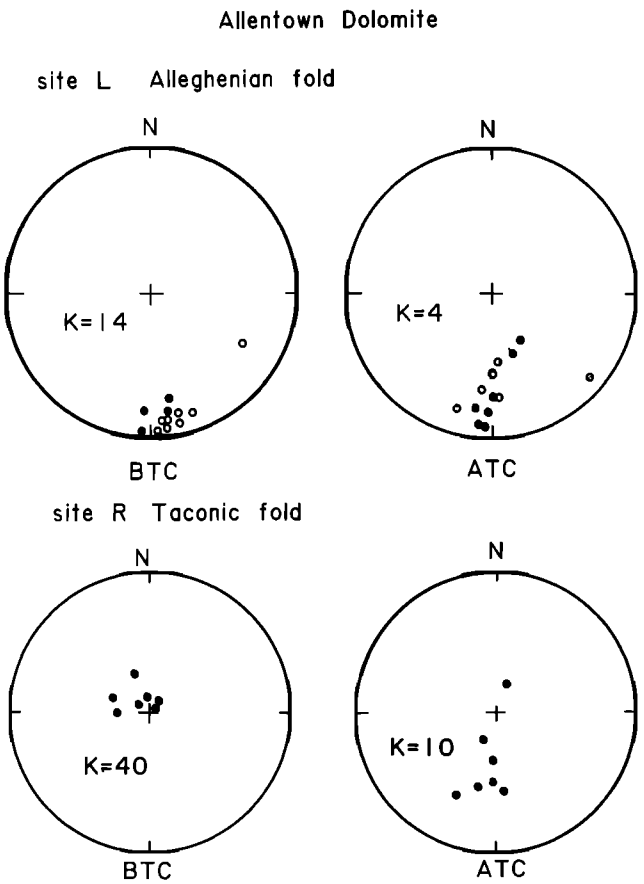



Fig. 9. Taconic and Alleghenian fold tests for two sites in the Allentown Dolomite. The Alleghenian fold test is at Lerch's Quarry, Bethlehem, PA (site L) and the Taconic fold test is at a quarry near Reigelsville, NJ (site R). Solid circles are lower hemisphere, open circles are upper hemisphere.

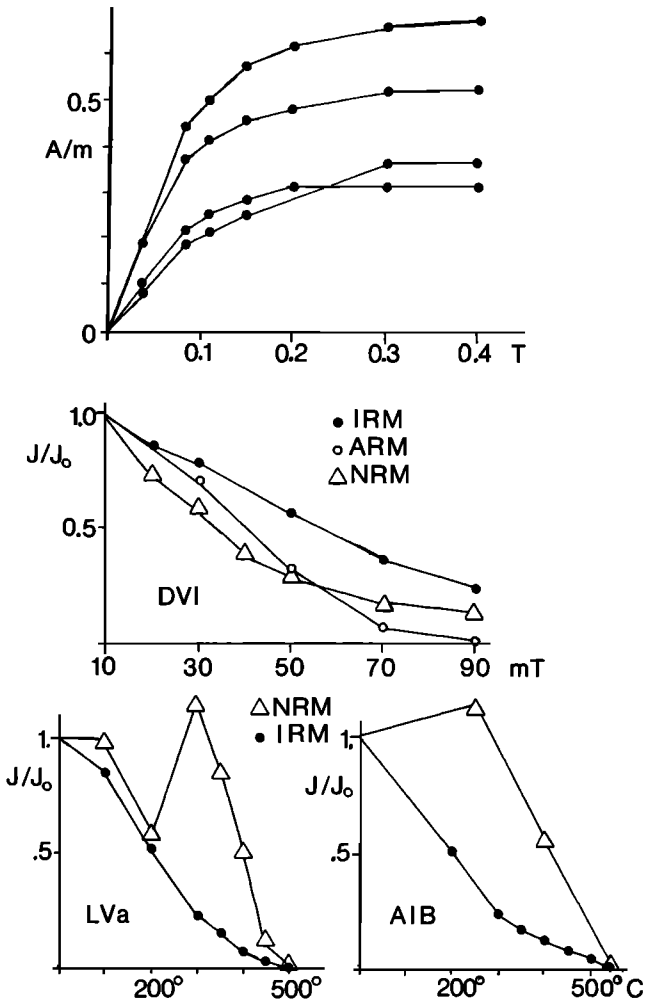



Fig. 10. IRM acquisition, alternating field demagnetization of NRM, ARM, and SIRM, and thermal degmanetization of NRM and SIRM for selected samples from the Allentown Dolomite and Leithsville Formation.

of I = -52° D = 26°. This is the most promising result of this study and will be pursued in a subsequent more detailed investigation. The preliminary Hardyston results suggest that a primary Cambrian magnetization may have been preserved which has been rotated clockwise by approximately 90° during either the Taconic or Alleghenian orogenies. It also may be possible that a Silurian secondary magnetization is recorded by these rocks. Unfortunately no fold tests could be applied to resolve this point.

The demagnetization behavior of the Leithsville and Allentown samples suggest that a recent, normal-polarity overprint is removed from the samples at low demagnetization steps. Following removal of this magnetization component a predominantly shallow and south magnetization is



Copyright American Geophysical Union

10.1029/GD012p0120. Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

isolated as the characteristic magnetization.

These magnetizations fail both Taconic and Alleghenian fold tests, and are probably Late Paleozoic in age, since the before tilt correction paleomagnetic pole for the Allentown lies at 46.6°N. lat.; 114.4°E. long. (Table 1), which is less than 10° from the North American Permian pole of Irving [1979]. In further support of this age interpretation is the observation that most of the magnetizations are reversed polarity before bedding tilt correction, the same polarity as the long reversed epoch in the Late Paleozoic [McElhinny, 1969]. In this case the magnetite which may be authigenic [Scotese et al., 1982] or primary magnetite was completely remagnetized as a result of deep burial and subsequent uplift and cooling associated with the Alleghenian orogeny.

Ordovician rocks in this part of Pennsylvania have conodont alteration indices (CAI) which range from 3.5-5 [Epstein, et al., 1977]. Epstein et al., [1977] suggest that the conodont color change is due to an increase in temperature accompanying the burial of the sediments. Since the Cambrian strata in eastern Pennsylvania were buried deeper than the Ordovician rocks, the conodont alteration indices set minimum temperatures for the Allentown and Leithsville formations of 200°C-300°C [Epstein et al., 1977]. This supports the suggestion that the secondary magnetizations of the Allentown and Leithsville formations are a thermal overprint.

It should be noted that the structural relationships which are interpreted to be synforms and antiforms by Drake [1967] could also be explained by imbricate thrusting. Since thrusting is observed to be a Taconic rather than an Alleghenian deformation behavior in this area, this would make the Alleghenian age fold test applied to sites Z, S, DL, and C actually a Taconic age "fold" test. To use this reasoning, however, would be in contradiction to Drake's interpretation of the area. In support of the approach used in this paper and Drake's interpretation, the symmetric anticline sampled at site L, which is interpreted to be an Alleghenian fold by Lash (personal communication) and Drake (personal communication), clearly gives a negative fold test for the Allentown Dolomite.

## Conclusions

The paleomagnetic results from the Cambrian Allentown dolomite and Leithsville formation indicate that these units were remagnetized during the Late Paleozoic. Results from the Allentown fail both Taconic and Alleghenian age fold tests. Leithsville data fail an Alleghenian age fold test. Comparison of the pole positions obtained from these formations with the North American APW path suggests that these units were remagnetized during the Early Permian. The preliminary data from the Cambrian Hardyston Quartzite are the most promising. Although no fold tests could be applied, the Hardyston may carry a primary remanence which has subsequently been rotated about a vertical axis.

Acknowledgments. The authors would like to thank V. A. Schmidt and D. V. Kent for the use of their paleomagnetic facilities and for helpful suggestions. We would also like to thank P. B. Myers and A. A. Drake for their assistance in locating sites and with interpretation of the geology. The comments of Chris Scotese, Rob Van der Voo and an anonymous reviewer were very helpful in improving this paper. This research was supported, in part, by a Lehigh University Forum Research Grant.

## References

Aaron, J. M., Petrology and origin of the Hardyston Quartzite (Lower Cambrian) in eastern Pennsylvania and western New Jersey, in Geology of Selected Areas in New Jersey and Eastern Pennsylvania, S. Subitzky, ed., Rutgers Univ. Press, New Brunswick, N.J. 21-34, 1969.

Creer, K. M., Paleozoic paleomagnetism, Nature, 219, 246-250, 1968.

Dankers, P. and P. LaPointe, Paleomagnetism of Lower Cambrian volcanics and a cross-cutting Cambro-Ordovician diabase dyke from Buckingham (Quebec), Can. J. Earth Sci., 18, 1174-1186, 1981.

Drake, A. A., Jr., Precambrian and Lower Paleozoic Geology of the Delaware Valley, New Jersey - Pennsylvania - in - Geology of Selected Areas in New Jersey and Eastern Pennsylvania, S. Subitzky, ed, Rutgers U. Press, New Brunswick, N.J. 51-131, 1969.

Drake, A. A., Jr., The Lyon Station - Paulins - Kill Nappe - The Frontal Structure of the Musconetcong Nappe System in Eastern Pennsylvania and New Jersey, U.S. Geological Survey, Prof. Paper 1023, 20pp., 1978.

Epstein, J. B. and A. G. Epstein, Geology in the region of the Delaware to Lehigh Water Gaps, Field Conference of Pennsylvania Geologists, 32nd, East Stroudsburg, Pennsylvania, Pennsylvania Geological Survey, Harrisburg, PA, 89 p., 1967.

Epstein, J. B. and Anita G. Epstein, Geology of the Valley and Ridge Province between Delaware Water Gap and Lehigh Gap, Pennsylvania; In: Geologyof Selected Areas in New Jersey and Eastern Pennsylvania, S. Subitzky, ed., Rutgers Univ. Press, New Brunswick, N.J., 132-205, 1969.

Epstein, A. G., J. B. Epstein, and L. D. Harris, Conodont color alteration - an index to organic metamorphism, U.S. Geological Survey Prof. Paper 995, U.S. Gov't. Printing Office, Washington, D.C., 27pp., 1977.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, J. Geophys. Res., 54, 131-167, 1949.



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Irving, E., Paleopoles and paleolatitudes of North America and speculations about displaced terrains, Can. J. Earth Sci., 16, 669-694, 1979.

Johnson, H. P., W. Lowrie, and D. V. Kent, Stability of Anhysteretic Remanent Magnetization in Fine and Coarse Magnetite and Maghemite Particles, Geophys. J. R. Astr. Soc., 41, 1-10, 1975.

Kent, D. V., Paleomagnetism of the Devonian Onondaga limestone revisited, J. Geophys. Res., 84, 3576-3588, 1979.

Lowrie, W. and Fuller, M., On the alternating field demagnetization characteristics of multidomain thermoremanent magnetization in magnetite, J. Geophys. Res., 76, 6339-6349, 1971.

Lowrie, W. and F. Heller, Magnetic properties of marine limestones, Rev. of Geophysics and Space Physics, 20, 171-192, 1982.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc., 8, 338-340, 1964.

McElhinny, M. W., The paleomagnetism of the Permian of southeast Australia and its significance regarding the problem of intercontinental correlation, Spec. Publ. Geol. Soc. Australia, 2, 61-67, 1969.

McElhinny, M. W. and W. D. Opdyke, Remagnetization hypothesis discounted: A paleomagnetic study of the Trenton limestone, New York State, Geol. Soc. Am. Bull., 84, 3697-3708, 1973.

Scotese, C., R. Van der Voo, and C. McCabe, Paleomagnetism of the Upper Silurian and Lower Devonian carbonates of New York State: Evidence for secondary magnetizations residing in magnetite, Phys. Earth Planet. Inter., 30, 385-395, 1982.

Sequin, M. K., K. V. Rao, and P. Arnal, Paleomagnetic study of Cambrian Potsdom Group sandstones, St. Lawrence Lowlands, Quebec, Earth Planet. Sci. Letters, 55, 433-449, 1981.

Turner, P., Depositional magnetization of carboniferous limestones from the Craven basin of northern England, Sedimentology, 22, 563-581, 1975.

Watts, D. R., R. Van der Voo, and S. C. Reeve, Cambrian paleomagnetism of the Llano Uplift, Texas, J. Geophys. Res., 85, 5316-5330, 1980.

Willard, B., Stratigraphy of the Cambrian sedimentary rocks of eastern Pennsylvania, Bull., Geol. Soc. of Am., 72, 1765-1776, 1961.

Zijderveld, J. D. A., AC demagnetization of rocks: Analysis of results in Methods in Paleomagnetism, ed by D. W. Collinson, K. M. Creer, and S. K. Runcorn, Elsevier, New York, 254-286, 1967.



Copyright American Geophysical Union

10.1029/GD012p0120, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GD012p0120 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License